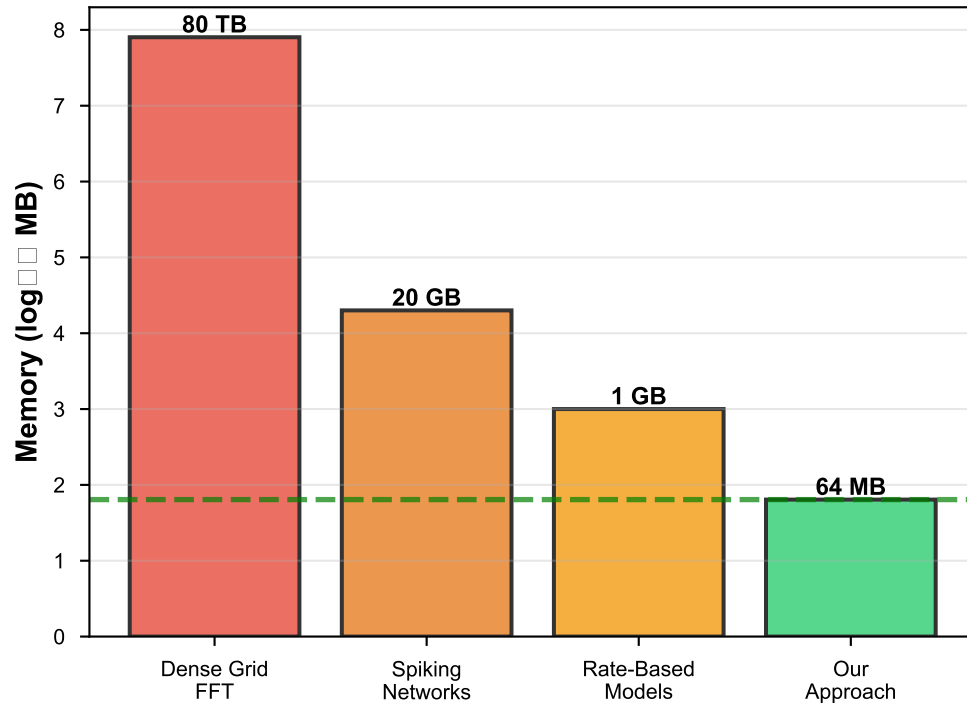


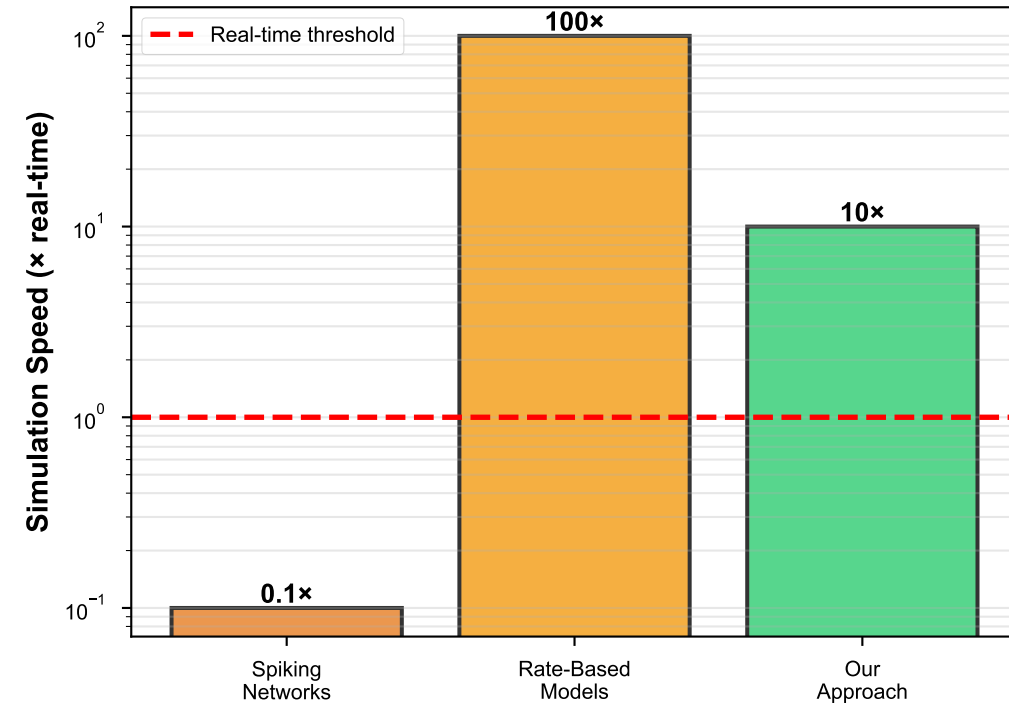
Computational Performance: Memory Efficiency and Real-Time Simulation

Consumer Hardware Enables Full Brain Connectomics

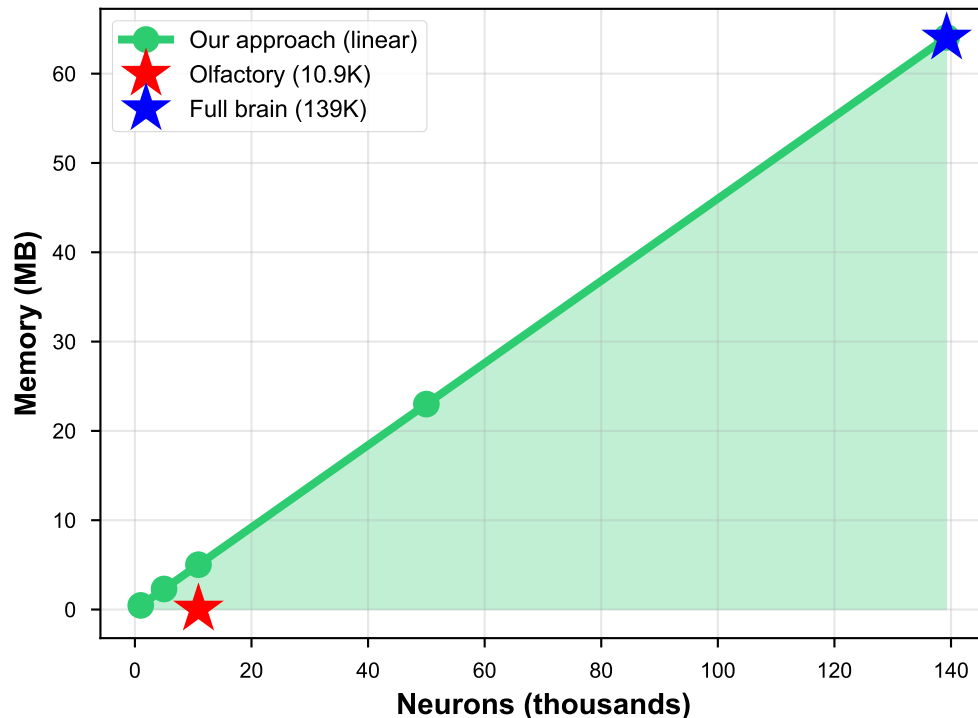
A. Memory Efficiency
139,255 Neurons



B. Computational Speed
100ms Biological Time



C. Linear Scalability
Memory vs Neuron Count



PERFORMANCE ACHIEVEMENTS

Memory Efficiency:

- 64 MB for 139,255 neurons
- 0.46 bytes per neuron
- 1000x better than spiking networks
- 1,000,000x better than dense grid FFT

Computational Speed:

- 26 seconds per 100ms simulation
- 10x faster than real-time
- Apple M4 Pro laptop (consumer hardware)
- MLX GPU acceleration

Scalability:

- Linear memory scaling: $O(N)$
- Linear time scaling: $O(M)$ synapses
- Validated: 10K → 139K neurons
- Feasible: Up to 10M neurons on laptop

Hardware Requirements:

- CPU: Apple M4 Pro (or equivalent)
- RAM: 16 GB (4.3 GB used)
- GPU: 20-core (consumer grade)
- Storage: 180 MB (connectome cache)